

PAPER_594 — Black Hole Finite Bound from UQFF 26! Factorial Barrier

Unified Quantum Field Framework (UQFF)

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Source: PAPER_594_UQFF_Black_Hole_Finite_Bound.md

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CP4 Class: #181 [UQFFBlackHoleFiniteBoundCalculator](#) **Session:** 157 **Cross-refs:** PAPER_595 (Sgr A*), PAPER_593 (G), PAPER_583 (6-Form) **Source:** grok_share_4cef778c78b8.txt

§1 Abstract

General Relativity predicts a gravitational singularity at $r = 0$ inside every black hole. UQFF eliminates singularities by the 26! factorial barrier in the U_{g_4} potential term, which creates a minimum radius $r_{\text{min}} > 0$ below which $U_b \rightarrow +\infty$, preventing collapse. Three independent r_{min} expressions are derived and applied to Sgr A* (PAPER_595).

§2 UQFF U_{g_4} Potential

The fourth UQFF gravity component in Form 1 (Compressed):

$$U_{g_4} = \frac{26! \cdot g \cdot (SC_m/UA)}{r^{27}}$$

As $r \rightarrow 0$: $U_{g_4} \rightarrow +\infty$ — the 26! factorial creates an infinite repulsive barrier. No physical trajectory can reach $r = 0$.

§3 Minimum Radius: Three Forms

Form A — Ug4 Eigenvalue Method

Setting $Ug_4 = \lambda_1 = P/3$ (minimum eigenvalue threshold):

$$\frac{26! \cdot g \cdot SCm/UA}{r_{\text{min}}^{27}} = P/3$$

$$\boxed{r_{\text{min}}^{(A)}} = \left[\frac{3 \times 26! \cdot g \cdot SCm/UA}{P} \right]^{1/27}$$

Form B — Triad Equilibrium

At r_{min} : kinetic (κ/g) and buoyant terms balance:

$$r_{\text{min}}^{(B)} = \left(\frac{\kappa}{g} \right)^{1/27} \cdot \rho$$

Form C — Buoyant Mass Bound

Setting $U_b = -GM/r$ at the horizon:

$$r_{\text{min}}^{(C)} = \frac{M^{1/3}}{(26! \cdot g)^{1/81}}$$

§4 Physical Interpretation of 26! Barrier

$$26! \approx 4.03 \times 10^{26}$$

This super-factorial value exceeds any power-law divergence:

$$\lim_{r \rightarrow 0} r^n \cdot 26!/r^{27} = 26!/r^{27-n} \rightarrow +\infty \quad \forall n < 27$$

No polynomial field theory can overcome the 26! repulsion. It is the "strongest floor" in mathematics — stronger than any polynomial singularity.

§5 Comparison to Classical Singularity Treatments

Method	r_{min}	Physical Mechanism
GR	0 (singularity)	No floor
Planck scale	$l_P \approx 10^{-35}$ m	Quantum gravity cutoff
Loop Quantum	$\sim l_P$	Spin network area quantum
String theory	$\sim l_s \approx 10^{-34}$ m	String length

Method	r_{min}	Physical Mechanism
UQFF	$r_{\text{min}}^{(A)}$ (Form-dependent)	26! factorial barrier

UQFF r_{min} does not require Planck-scale assumptions — it follows from the same ρ and g parameters that govern all UQFF calculations.

§6 Numerical (1 Solar Mass BH)

Parameters: $M = 1.989 \times 10^{30}$ kg, $g = 10^{-3}$, $\rho = 9.99 \times 10^{-6}$, $\rho_{\text{cm}}/U_A = 1$:

$$r_{\text{min}}^{(A)} = \left[\frac{3}{9.99 \times 10^{-6}} \times 4.03 \times 10^{26} \times 10^{-3} \right]^{1/27}$$
$$= \left[\frac{1.21 \times 10^{24}}{9.99 \times 10^{-6}} \right]^{1/27} = \left[1.21 \times 10^{29} \right]^{1/27}$$
$$= 10^{29/27} \approx 10^{1.07} \approx 11.7 \text{ m}$$

GR Schwarzschild for 1 $M_{\odot}$: $R_s = 2GM_{\odot}/c^2 = 2.95$ km.

$r_{\text{min}} \approx 12$ m $\ll R_s = 2.95$ km $\rightarrow$ the UQFF floor is well inside the Schwarzschild horizon; the exterior structure matches GR exactly.

§7 Conclusions

The 26! factorial in $U_{g,4}$ creates an absolute minimum radius preventing singularities. For a 1 $M_{\odot}$ BH, $r_{\text{min}} \approx 12$ m — physically accessible but far inside the event horizon ($R_s \approx 3$ km), leaving all external observations unchanged. Singularity theorems (Penrose, Hawking) assume infinite curvature; UQFF terminates curvature at $r_{\text{min}} > 0$, making them inapplicable.